

Question Paper

Exam Date & Time: 27-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

REINFORCEMENT LEARNING [CSE_4028]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Develop the interaction among the elements of Reinforcement Learning by applying them to a self-driving car learning to follow traffic rules. (5)

A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Model how the Magic Square representation can be used to model and solve the Tic-Tac-Toe problem in Reinforcement Learning, highlighting how this approach simplifies the detection of winning conditions and state evaluation. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Identify practical examples where Reinforcement Learning can be applied to enable systems to learn optimal behavior through interaction with their environment. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Analyze whether the following probability assignments satisfy the axioms of probability. A coin is tossed once, and the probabilities are assigned as: $P(\text{Head}) = 0.6$, $P(\text{Tail}) = 0.5$. Do these probabilities form a valid probability model according to the axioms of probability? (5)

A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) A fair coin is tossed three times, and let X be defined as the number of tails observed. Solve for range R_X and derive the probability mass function $P_X(x)$. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Identify how the Cumulative Distribution Function (CDF) can be used to determine the probability that a continuous random variable takes a value less than or equal to a given number. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Analyze how Action-Value Methods can be modified to effectively track a nonstationary problem in the n-Armed Bandit scenario, using a constant step-size parameter to adapt to changing action values. (5)
- A)
 Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Apply the unified notation for episodic and continuing tasks can be applied in a real-world scenario such as a robot navigating a warehouse, where some tasks have a clear endpoint (like delivering a package) while others run continuously (like monitoring shelf inventory). Describe how the discount factor and reward formulation adapt to both cases. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) Identify how Upper-Confidence-Bound (UCB) methods can be utilized to improve action selection by considering both estimated rewards and uncertainty. (2)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Identify how optimal value functions can be applied in a game-playing scenario such as chess or tic-tac-toe, to enable the agent to evaluate possible future states and select actions that maximize long-term rewards (5)
- A)
 Course Outcome: "CO4" / Cognitive Level: "Application"
- B) Compare Policy Evaluation and Policy Improvement in Dynamic Programming, highlighting how each contributes to the process of finding the optimal policy in a Markov Decision Process. (3)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- C) Identify how goals and reward signals influence the agent's behavior in a finite Markov Decision Process. (2)
- Course Outcome: "CO4" / Cognitive Level: "Application"
- 5) Examine how the SARSA algorithm updates action values through on-policy learning, and analyze how its behavior differs from Q-learning in terms of policy evaluation and exploration-exploitation balance (5)
- A)
 Course Outcome: "CO5" / Cognitive Level: "Analysis"
- B) Apply R-Learning in an average-reward setting, where the agent learns to optimize long-term average performance instead of discounted rewards, particularly in continuing tasks like industrial process control or network routing. (3)
- Course Outcome: "CO5" / Cognitive Level: "Application"
- C) Apply eligibility traces in an Actor-Critic framework to update both the actor and critic more efficiently during continuous interaction with the environment. (2)
- Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	10

2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10

Total 50

* **NA** - CO Not Selected

Event Name	Subject
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LEARNING

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	11	32(64%)
4	Analysis	4	18(36%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)